AI Healthcare Assistant

Run Manual: commands to install, start and stop the project

Windows (PowerShell / Command Prompt) · macOS / Linux · VS Code

What you need

Requirement	Details
Python 3.10 or newer	Check with `python --version`. Tested with Python 3.12.
Internet (first time only)	Needed once to download the packages in requirements.txt. The app itself runs offline.
A web browser	Chrome, Edge or Firefox. The app opens at http://localhost:8501.

Quick start (Windows, after the first setup)

If you have already done the first-time setup below, these three commands are all you need each time:

```
cd "C:\Users\LENOVO\OneDrive\Desktop\Project 3 Predictive Healthcare Resource
Planning – Alternative Stack\Project 3 Mock (Healthcare)"
.venv\Scripts\activate
streamlit run app.py
```

Keep the quotes around the folder path: it contains spaces and a dash character. A long path wraps onto two lines here; type it as one line.

First-time setup and run (Windows PowerShell)

Step 1. Open a terminal in the project folder

In VS Code choose *File* → *Open Folder* and select the project folder, then press **Ctrl +** ` to open a terminal. Or, in File Explorer, open the project folder, click the address bar, type **powershell** and press Enter. Or move into the folder from any PowerShell window:

```
cd "C:\Users\LENOVO\OneDrive\Desktop\Project 3 Predictive Healthcare Resource
Planning – Alternative Stack\Project 3 Mock (Healthcare)"
```

A long path wraps onto two lines here; type it as one line.

Step 2. Create a virtual environment

This keeps the project's packages separate from the rest of your computer. Run it once.

```
python -m venv .venv
```

Step 3. Activate the virtual environment

Run this every time you open a new terminal. You will see `(.venv)` at the start of the line.

```
.venv\Scripts\activate
```

If PowerShell says running scripts is disabled, run this once and try again: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

Step 4. Install the packages

Downloads Streamlit, pandas, scikit-learn and the other libraries. Run it once (or again if requirements.txt changes).

```
pip install -r requirements.txt
```

Step 5. Start the app

The browser opens automatically. If it does not, open **http://localhost:8501** yourself.

```
streamlit run app.py
```

Step 6. Stop the app

Click in the terminal where the app is running and press **Ctrl + C**. To leave the virtual environment afterwards:

```
deactivate
```

Windows Command Prompt (cmd)

The same steps work in Command Prompt. Only the activate command is different:

```
cd /d "C:\Users\LENOVO\OneDrive\Desktop\Project 3 Predictive Healthcare Resource
Planning — Alternative Stack\Project 3 Mock (Healthcare)"
python -m venv .venv
.venv\Scripts\activate.bat
pip install -r requirements.txt
streamlit run app.py
```

macOS / Linux

```
cd "/path/to/Project 3 Mock (Healthcare)"
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py
```

Other useful commands

Task	Command
Run the safety tests	`python test_engine.py` Expected output: *All safety tests passed*
Use a different port	`streamlit run app.py --server.port 8502`
Start without opening a browser	`streamlit run app.py --server.headless true`
Open on a phone (same Wi-Fi)	Use the *Network URL* printed in the terminal, e.g. http://192.168.x.x:8501
If 'streamlit' is not recognized	`python -m streamlit run app.py`
Clear saved history	Use *Delete all records* in the History tab, or delete the file `history.db` while the app is stopped.

Troubleshooting

Problem	Fix
Port 8501 is not available	The app is already running in another terminal. Open http://localhost:8501, stop the other one with Ctrl + C, or use a different port.
'python' is not recognized	Install Python from python.org and tick *Add python.exe to PATH*, or use `py` instead of `python`.
ModuleNotFoundError	The virtual environment is not active or packages are missing. Run step 3, then step 4.
Scanned PDFs give no text	Optional OCR needs `pip install pytesseract pdf2image` plus the Tesseract and Poppler programs.
Page looks unstyled or old	Refresh the browser (Ctrl + F5). Fonts load from Google Fonts when online; the app still works offline with a fallback font.
Error after typing "chest pain" in the description	A known bug in engine.py. Pick *Chest pain* from the symptom list instead until it is fixed.

Project files

File	Purpose
app.py	Streamlit interface and results dashboard
engine.py	Validation, text understanding, red flags, risk levels, conditions, medicines
ml.py	Swappable ML models for the second opinion
knowledge.py	PDF/text upload and document search (sources)
datasets.py	CSV/Excel/JSON profiling, quality checks, test training
storage.py / history.db	Local SQLite history (stays on this computer)
.streamlit/config.toml	Colour theme for the interface
requirements.txt	List of Python packages to install
test_engine.py	Quick safety tests

This tool gives general health information and decision support only. It is not a diagnosis and does not replace a doctor. In an emergency call your local emergency number (1122 in Punjab, Pakistan; 911, 999 or 112 elsewhere).